# UMANG P. KAKLOTAR

**AI/ML Engineer**

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Portfolio: https://protfolio-frontend.upkaklotar7.workers.dev | GitHub: https://github.com/umangpkaklotar

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## PROFESSIONAL SUMMARY

Entry-level AI/ML Engineer with practical software development experience and hands-on expertise building AI-powered applications using Python, Machine Learning, Computer Vision, and Generative AI. Experienced in developing custom YOLOv8 object detection pipelines, real-time optical inspection (AOI), ROS 2 and Gazebo robotics simulations, and Retrieval-Augmented Generation (RAG) platforms using Gemini embeddings, FAISS vector search, and Groq LPU inference. Solid engineering foundation with experience in application development, REST API integration, and cloud deployment. Currently attending AI/ML classes at DataMites in Bangalore.

## TECHNICAL SKILLS

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| **Programming Languages** | Python, JavaScript (ES6+), PHP, HTML5, CSS3, SQL |
| **Machine Learning** | scikit-learn, Pandas, NumPy, Feature Engineering, Model Training, EDA, Regression |
| **Deep Learning & CV** | Ultralytics YOLOv8, PyTorch, Torchvision, OpenCV (cv2), Scikit-Image, Object Detection |
| **Generative AI & LLMs** | RAG, LLM Integration, Google Gemini, Groq LPU, LangChain, Prompt Engineering, Citation Grounding |
| **Vector Search & Retrieval** | FAISS (IndexFlatIP), Vector Embeddings (Gemini Embedding-2), Semantic Search, Document Chunking |
| **Robotics & Simulation** | ROS 2 (Jazzy), Gazebo Sim 8, RViz2, Conveyor Automation, CvBridge, Telemetry |
| **Backend & Deployment** | FastAPI, Uvicorn, REST APIs, WebSockets, Server-Sent Events (SSE), Docker, Git, GitHub |
| **Web Technologies** | React.js, Tailwind CSS, Bootstrap, JavaScript, Authentication (Clerk), Role-Based Access Control |
| **Databases** | MongoDB, MySQL |

## PROJECTS

**PCB Defect Detection: AI-Powered Industrial Inspection Platform** | Jan 2025 – Present

Live Demo: https://pcb-frontend.getvoroa.com/

- Engineered an industrial AI inspection pipeline using custom-trained Ultralytics YOLOv8 (best.pt) detecting 7 defect classes (Copper, Mousebite, Open, Pin-hole, Short, Spur, Missing Hole) with high confidence.
- Developed real-time inspection supporting desktop uploads and shop-floor mobile camera streams with automated Pass / Rework / Scrap industrial quality decisions.
- Built virtual smart manufacturing digital twin in Gazebo Sim 8 with ROS 2 conveyor automation; integrated virtual proximity sensors and virtual camera feeds via CvBridge for automated zero-contact defect detection.
- Developed asynchronous FastAPI backend with MongoDB recording PCB IDs, defect coordinates, and machine status for historical quality auditing.

**Document AI: Enterprise Multi-Format RAG Platform** | Jan 2025 – Present

Live Demo: https://rag-frontend-omega-three.vercel.app/

- Engineered a Retrieval-Augmented Generation (RAG) platform supporting multi-format ingestion for PDF, DOCX, CSV, TXT, and Markdown documents with semantic overlap chunking.
- Generated batch vector embeddings using Google Gemini Embedding-2, indexed into in-memory FAISS IndexFlatIP for sub-millisecond similarity retrieval.
- Integrated Groq LPU inference delivering real-time streaming answers via Server-Sent Events (SSE) with verified source citation bubbles and anti-hallucination prompt guardrails.
- Implemented multi-tenant user isolation with Clerk JWT authentication and encrypted MongoDB conversation history.

**Real Estate Valuation: Home Price Prediction System** | 2025

- Conducted comprehensive Exploratory Data Analysis (EDA) on multi-feature housing datasets with automated outlier filtering, categorical encoding, and feature scaling.
- Trained linear and ensemble regression models using Scikit-Learn with cross-validation and hyperparameter tuning, achieving an R2 score of 0.942.

## PROFESSIONAL EXPERIENCE

**Frontend Developer Intern, Jemistry Info Solutions LLP, Surat, India** | 6 Months (2025)

- Developed responsive, accessible web interfaces converting Figma UI/UX designs into production-ready React and Bootstrap components.
- Integrated RESTful APIs using Axios and Fetch API; implemented client-side state, error handling, and payload validation.
- Optimized frontend rendering performance, cross-browser compatibility, and modular styling across desktop and mobile devices.
- Collaborated in an agile team workflow using Git and GitHub for pull requests, code reviews, and structured version control.

## EDUCATION

**Bachelor of Computer Applications (BCA), Smt ZS Patel College, Surat** | Jun 2022 – 2025

Relevant Coursework: Computer Vision, Machine Learning, DSA, OOP, DBMS, Web Technologies, Operating Systems

**AI/ML Classes — DataMites, Bangalore, India** | Present

Currently attending AI/ML training focusing on building practical knowledge and skills in Artificial Intelligence and Machine Learning.